

HONG WANG

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RESEARCH INTERESTS

- Area: AI, Augmented Reality (AR), Computer Vision (CV), Large Language Models (LLMs)

EDUCATION

8/2023 - Present	University Of South Florida Ph.D. in Computer Science Advisor: Prof. Zhao Han	Tampa, FL, USA
2/2021 - 12/2022	Stony Brook University M.S. in Computer Science	Stony Brook, NY, USA
9/2016 - 6/2020	Hohai University B.S. in Information and Computing Science	Nanjing, Jiangsu, China

WORKING EXPERIENCE

8/2023 – Present	Research Assistant • Work on interactive robot systems involving LLMs, AR interfaces, and projector-camera perception. • Designed and evaluated AR indicators for robot field-of-view communication. • Continually contributed to user studies and paper writing. • Developed real-time robot behavior pipelines integrating speech, gestures, and tablet UI for Misty robot. • Mentored undergraduate researchers, supporting project planning, experiment design, and problem-solving.	Reality, Autonomy, and Robot Experience (RARE) Lab
6/2024 - 8/2024	Voice AI Summer School Researcher • Trained a CNN model using MFCC features, achieving 80.85% accuracy on labeled data. • Improved model performance by 26.4% (from 65.1% to 91.5%) via data augmentation and pre-trained models (ResNet50, EfficientNet B0).	University of South Florida
1/2023 – 6/2023	Software Developer • Developed Java-based web applications using Spring, Hibernate, and MySQL, including multi-table database schemas and REST APIs for data integration. • Participated in Agile development with responsibilities in code reviews, testing, and documentation.	Best High Technologies, Princeton, NJ, USA

PUBLICATIONS

- **Hong Wang**, Ngoc Bao Dinh, and Zhao Han. "Evaluating Dynamic Surface Compensation for Robots with Projected AR," *HRI 2026 Late-Breaking Reports*
- **Hong Wang**, Ridhima Phatak, James Ocampo, and Zhao Han. "Indicating Robot Vision Capabilities with Augmented Reality," *International Journal of Social Robotics*.
- **Hong Wang**, Zhao Han. "Towards Robot Persuasiveness Self-Assessment for LLM-Powered Recipe Recommender Robot," *Under review by RSS*
- Xiangfei Kong, Rex Gonzalez, Nicolas Echeverria, **Hong Wang et al.**. "Bloom Preview: A Low-Cost LLM-Powered Social Robot" *HRI 2026 Interactivity*
- **Hong Wang**, Maria Julia Vidal, and Zhao Han. "Exploring Familiar Design Strategies to Explain Robot Vision Capabilities," *Explainability for Human-Robot Collaboration: Real-World Concerns Workshop (X-HRI) at HRI 2025, 2025*
- Adrian Lozada, Uthman Tijani, Villa Keth, **Hong Wang**, and Zhao Han. "Anywhere Projected AR for Robot Communication: A Mid-Air Fog Screen-Robot System," *2025 ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2025*.
- **Hong Wang**, Tam Do, and Zhao Han. "To Understand Indicators of Robots' Vision Capabilities," *VAM-HRI 2024 — 2024 International Workshop on Virtual, Augmented, and Mixed-Reality for Human-Robot Interactions at HRI 2024, 2024*
- **Hong Wang**, Tam Do, and Zhao Han. "Designing Indicators to Show a Robot's Physical Vision Capability," *IEEE VR 2024 Poster — The 31st IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (IEEE VR), 2024*
- Uthman Tijani, **Hong Wang**, and Zhao Han. "Towards Reproducible Language-Based HRI Experiments: Open-Sourcing a Generalized Experiment Project," *2024 ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2024*.

SELECTED APPLIED PROJECTS

2025 – Present	LLM-Powered Persuasive Social Robot for Healthy Meal Decisions (Misty) • Developed an LLM-powered Misty robot system for healthy recipe recommendation with synchronized spoken dialogue and tablet-based multimodal interaction. • Designed and evaluated five persuasive strategies across five LLMs, generating 2,500 single-turn recommendations and 1,000 multi-turn persuasive dialogues. • Built a real-time multimodal interaction pipeline coordinating speech, facial expressions, gestures, LED indicators, and tablet UI on the Misty robot. • Developed an automated evaluation framework to compare persuasion effectiveness across multiple LLMs (GPT-4o, Claude, Gemini, LLaMA, DeepSeek), enabling rapid iteration and design comparison.
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- 2024 – Present **Projected Augmented Reality Display System for Mobile Robots (Fetch, Unitree Go2)**
- Built an end-to-end **projected AR communication system** enabling mobile robots to convey spatial information through mid-air visual displays.
 - Integrated custom hardware and software components, including **ROS-based control**, **Arduino-driven actuation**, and real-time synchronization with robot navigation.
 - Iteratively refined system design through rapid experimentation, achieving a **60% increase in effective display area** and improved deployment robustness.
- 2023 – 2024 **Mixed Reality Interfaces for Human–Robot Collaboration (Pepper, HoloLens 2)**
- Designed and implemented **AR visualization interfaces in Unity/C#** to communicate robot field-of-view and perceptual capabilities to human collaborators.
 - Prototyped **multiple interface paradigms** (egocentric and allocentric) and evaluated their impact on user understanding, confidence, and cognitive load.
 - Conducted a controlled user study (**N=41**) with Bayesian statistical analysis, translating findings into **actionable design guidelines** for transparent human–robot interaction.

PROFESSIONAL SERVICE

- **Reviewer**
 - IEEE International Conference on Robotics and Automation (ICRA) 2026.
 - Human-Robot Interaction (HRI) Conference 2025, 2026.
 - Transactions on Human-Robot Interaction (THRI) 2025.
 - IEEE RO-MAN Conference 2024, 2026.
- **Student Volunteer of IEEE VR 2024 Conference**
- **Student Volunteer of Human-Robot Interaction 2024 Conference**

PROFESSIONAL DEVELOPMENT

- **UR2PhD Graduate Student Mentor Training** | Computing Research Association | 01/2025 – 04/2025
- **2024 CRA-WP Grad Cohort Workshop for Women** | Computing Research Association | 4/2024
- **Staff Member, USF Human-Centered AI and Robotics Consortium (HARC)** | University of South Florida | 2024–Present
- **Bridge2AI Voice Scholar** | Bridge2AI Voice Consortium | 2024

HONORS AND GRANTS

- **Winner of USF Voice AI Summer School Hackathon** | Bridge2AI Voice | 08/2024
- **USF Engineering Alumni Society Conference Grant** | \$500 | 05/2024, 03/2025
- **NSF I-Corps Site subgrant** | Entrepreneur Lead | \$3,000 | 1/2024–3/2024

LEADERSHIP & MENTORING

- 2023 – Present **Teaching Assistant & Research Mentor** **University of South Florida**
- TA:** COP 4931 Robotics Process Automation, CIS 6930 Augmented Reality, CEN 4020 Software Engineering, COP 4710 Database Design
- Mentoring:** Supervised over 30 undergraduate researchers in Python, Unity/MRTK, HoloLens 2 development, and user study design for HRI projects

OUTREACH AND ENGAGEMENT

- 10/05/2024 **DevFest 2024** **University of South Florida**
- Presented a live demo of the Pepper robot and HoloLens 2 to over 30 attendees including students and industry professionals from tech companies.
- 03/01/2024 **USF Innovation Fest** **University of South Florida**
- Showcased research on indicating a robot's vision capability with a Pepper robot and HoloLens 2, engaging over 20 undergraduate and graduate students.

TECHNICAL SKILLS

Programming: Python, C++, C#, Java, JavaScript (React), SQL, R
Machine Learning & NLP: Deep Learning, PyTorch, Hugging Face Transformers, TensorFlow, BERTopic
Cloud & Systems: Docker, Google Cloud, AWS, Azure
Robotics Frameworks: ROS (ROS1/ROS2)
AR/VR Development: Unity, Vuforia, MRTK

Simulation Tools: Rviz, Webots, Gazebo
3D & Graphics: Blender
Data & Visualization: Pandas, NumPy, Matplotlib, D3.js, Tableau
Collaboration & Tools: Git, VS Code, Overleaf, Google Colab
Languages: English (fluent), Mandarin (native)